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CBCS SCHEME

USN ☐☐☐☐☐☐☐☐☐☐

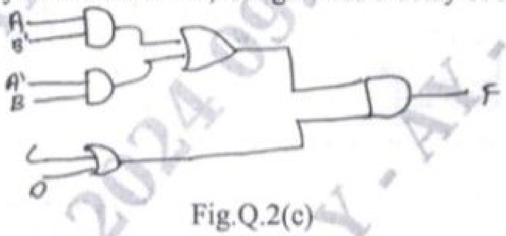
BCS302

Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

Digital Design and Computer Organization

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	**M**	**L**	**C**
Q.1	**a.**	Obtain a minimum product of sums with a Karnaugh map. $F(w, x, y, z) = x'z' + wyz + w'y'z' + x'y$.	**10**	**L3**	**CO1**
	b.	Find the minimum sum of products for each function using a Karnaugh map i) $F_1(a, b, c) = M_0 + M_2 + M_5 + M_6$ ii) $F_2(d, e, f) = \Sigma m(0, 1, 2, 4)$ iii) $F_3(r, s, t) = rt' + r's' + r's$	**10**	**L3**	**CO1**
		OR			
Q.2	**a.**	Identify the prime implicants and essential prime implicants of the following functions: i) $f(A, B, C, D) = \Sigma(1, 3, 4, 5, 10, 11, 12, 13, 14, 15)$ ii) $f(W, X, Y, Z) = \Sigma(0, 1, 2, 5, 7, 8, 10, 15)$.	**10**	**L3**	**CO1**
	b.	Write the verilog code for the given expression using dataflow and behavioral model where $Y = (AB' + A'B)(CB + AD)(AB'C + AC)$.	**5**	**L2**	**CO1**
	c.	Write the verilog code and time diagram for the given circuit with propagation delay where the AND, OR gate has a delay of 30ns and 10ns.  Fig.Q.2(c)	**5**	**L2**	**CO1**
		Module – 2			
Q.3	**a.**	What is Latch? With neat diagram, explain S-R latch using NOR gate. Derive characteristics equation.	**10**	**L3**	**CO2**
	b.	What is priority encoder? Design 4:2 priority encoder with necessary diagrams.	**10**	**L3**	**CO2**
		OR			
Q.4	**a.**	Design and explain four bit adder with carry look ahead.	**10**	**L3**	**CO2**
	b.	What is multiplexer? Design 9:1 mux using 2:1 mux.	**10**	**L3**	**CO2**

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		Module – 3			
Q.5	**a.**	Explain four types of operation performed by computer with an example.	10	L2	CO3
	b.	Show how below expression will be executed in one address, two address zero address and three address processor in an accumulator organization $X = (A * B) + (C * D)$.	10	L1	CO3
		OR			
Q.6	**a.**	What is addressing mode? Explain different types of addressing mode with an examples.	10	L2	CO3
	b.	With a neat diagram, explain basic operational concepts of a computer.	10	L2	CO3
		Module – 4			
Q.7	**a.**	Explain the following with respect to interrupts with diagram. i) Vector interrupt ii) Interrupt nesting iii) Simultaneous request.	10	L2	CO3
	b.	Explain Direct Memory Access with a neat diagram.	10	L2	CO3
		OR			
Q.8	**a.**	What is Bus arbitration? Explain different types of bus arbitration.	10	L2	CO3
	b.	Discuss different types of mapping functions of coaches.	10	L2	CO3
		Module – 5			
Q.9	**a.**	Draw and explain the single-bus organization of the data path inside a processor.	10	L2	CO4
	b.	List out the actions needed to execute the instruction ADD (R3), R1 write and explain the sequence of control steps for the execution of the same.	10	L2	CO4
		OR			
Q.10	**a.**	Analyze how does execution of a complete instruction carry out.	10	L4	CO4
	b.	What is pipeline? Explain the performance of pipeline with an example.	10	L4	CO4

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CBCS SCHEME - Make-Up Exam

USN

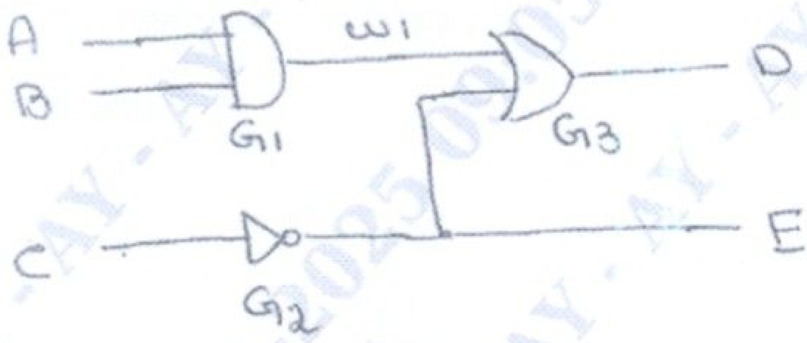
BCS302

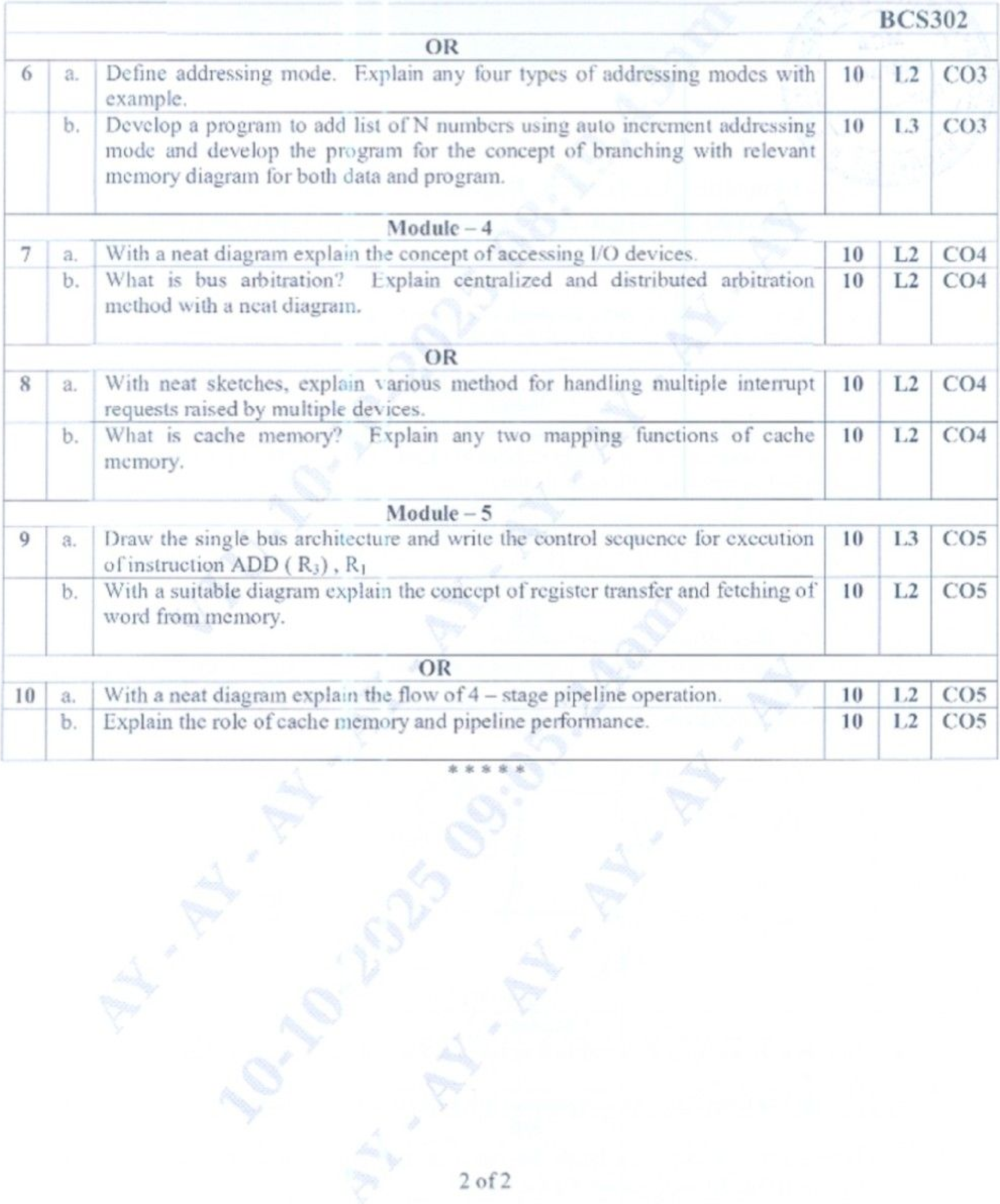
Third Semester B.E/B.Tech. Degree Examination, June/July 2025

Digital Design and Computer Organization

Time: 3 hrs. Max. Marks:100

Note: *1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
1	a.	Apply the basic theorems and postulates of Boolean algebra to simplify the following expression to minimum number of literals and draw the original and simplified expressions with logic diagram. i) $xy + x^{1}z + yz$ ii) $x + x^{1}y$	10	L3	CO1
	b.	Apply k – map Technique to simplify the function $F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$ and $d(w, x, y, z) = \sum(0, 2, 5)$ and also implement the simplified function with NAND gate.	10	L3	CO1
		OR			
2	a.	Describe Map Method for three variables.	7	L2	CO1
	b.	Explain Positive and Negative logic and prove that positive AND is equal to negative OR.	7	L2	CO1
	c.	Develop a verilog gate –level description of the circuit shown in Fig. Q. 2 (c)  Fig. Q. 2(c)	6	L3	CO1
		Module – 2			
3	a.	Design a combinational circuit that converts a four – bit BCD code to Excess 3 code.	10	L2	CO3
	b.	Design 3 to 8 decoder. Also implement full adder using 3 to 8 decoder.	10	L3	CO2
		OR			
4	a.	Describe 4 × 1 mox with block diagram and truth table. Also develop a behavioral model verilog code for 4 × 1 mux.	10	L3	CO2
	b.	What is flip flop? Explain positive edge triggered D type of flip flop.	10	L2	CO2
		Module – 3			
5	a.	With a neat block diagram explain the functional units of computer.	10	L2	CO3
	b.	Explain the Basic Instruction types with examples.	10	L2	CO3



		OR			
6	a.	Define addressing mode. Explain any four types of addressing modes with example.	10	L2	CO3
	b.	Develop a program to add list of N numbers using auto increment addressing mode and develop the program for the concept of branching with relevant memory diagram for both data and program.	10	L3	CO3
		Module – 4			
7	a.	With a neat diagram explain the concept of accessing I/O devices.	10	L2	CO4
	b.	What is bus arbitration? Explain centralized and distributed arbitration method with a neat diagram.	10	L2	CO4
		OR			
8	a.	With neat sketches, explain various method for handling multiple interrupt requests raised by multiple devices.	10	L2	CO4
	b.	What is cache memory? Explain any two mapping functions of cache memory.	10	L2	CO4
		Module – 5			
9	a.	Draw the single bus architecture and write the control sequence for execution of instruction ADD (R_3) , R_1	10	L3	CO5
	b.	With a suitable diagram explain the concept of register transfer and fetching of word from memory.	10	L2	CO5
		OR			
10	a.	With a neat diagram explain the flow of 4 – stage pipeline operation.	10	L2	CO5
	b.	Explain the role of cache memory and pipeline performance.	10	L2	CO5

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BCS302

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BANGALORE

Third Semester B.E./B.Tech. Degree Examination, June/July 2025

Digital Design and Computer Organization

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Demonstrate the positive and negative logic using AND gate.	05	L2	CO1
	b.	Show that NAND and NOR function are commutative and but not associative.	05	L3	CO1
	c.	Simplify the Boolean function $F(w,x,y,z) = \Sigma m\,(0,1,2,4,6,7,9,12,14)$ using K-Map and implement using NAND gates.	10	L3	CO1
		OR			
Q.2	a.	What is Binary Logic? List out any 4 laws of logic.	05	L1	CO1
	b.	Find the POS expression for $F(A.B.C.D) = \prod M(2,3,5,8,10,13,14) + d(1,6,7,11)$ and realize it using NOR gates.	05	L3	CO1
	c.	Simplify the following Boolean functions using K-map and write the verilog program for realizing the minimized expression. i) $F(X,Y,Z) = \Sigma m\,(0,1,4,5,6) + d\,(2,3)$ ii) $F(W,X,Y,Z) = \Sigma m\,(5,6,7,12,14,15) + d(9,11,13)$	10	L3	CO1
		Module – 2			
Q.3	a.	Define Decoder Implement the following boolean functions using a decoder $F_1(A,B,C) = \Sigma m(1,3,4,7)$ $F_2(A,B,C) = \Sigma m(0,2,,3,6)$	06	L3	CO2
	b.	Write the verilog program to implement full adder and full sub tractor circuits.	06	L3	CO2
	c.	Design an octal to Binary Encoder.	08	L3	CO2
		OR			
Q.4	a.	Define Multiplexer. Implement the Boolean function $F\,(A,B,C,D) = \Sigma m(1,3,4,11,12,13,14,15)$ with 8:1 multiplexer.	06	L3	CO2
	b.	Explain the working of 4 – bit adder using 4 full adders.	06	L2	CO2
	c.	Write the characteristic equation, Excitation table and FSM representations for SR,JK and D flip flops	08	L2	CO2
		Module – 3			
Q.5	a.	With a block diagram explain the processor and Memory communication.	06	L2	CO3
	b.	With relevant examples, Explain the following addressing modes. i) Index ii) Base with index and offset iii) Indirect.	06	L2	CO3
	c.	Demonstrate the instruction execution and sequencing for $C \leftarrow [A] + [B]$	08	L2	CO3

		OR			
Q.6	**a.**	Describe the Big Endian and little endian address assignment with examples	**06**	L2	CO3
	b.	The Registers R1 and R2 has decimal values 1200 and 4600. Calculate the EA of the memory operand in each of the following instructions when they are executed in sequence. i) load 20 (R1), R5 ii) Move # 3000, R5 iii) Store R5, 30(R1,R2) iv) add – (R2), R5 v) Sub (R1)+, R5 vi) add (R2)+,R1	**06**	L3	CO3
	c.	Demonstrate the Branching operations using a loop to add n – numbers with block diagram.	**08**	L3	CO3
		Module – 4			
Q.7	**a.**	Explain the effect of size, cost and speed in Memory hierarchy.	**10**	L2	CO4
	b.	Explain Centralized and distributed Bus arbitration approaches.	**10**	L2	CO4
		OR			
Q.8	**a.**	Describe the different memory mapping functions.	**10**	L2	CO4
	b.	Explain how to handle interrupt from multiple devices using daisy chain and priority scheme.	**10**	L2	CO4
		Module – 5			
Q.9	**a.**	With a neat diagram, explain the single bus organization of the data path inside a processor.	**10**	L2	CO5
	b.	Describe in detail, the basic idea of instruction pipeline.	**10**	L2	CO5
		OR			
Q.10	**a.**	Explain the complete set of operations involved in executing the instruction ADD (R3), R1 along with control sequence.	**10**	L2	CO5
	b.	Explain the process of storing a word from processor to memory.	**10**	L2	CO1

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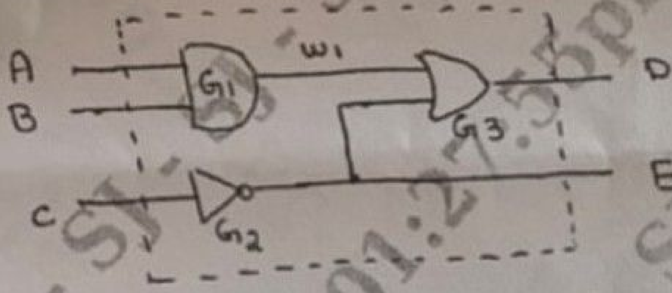
Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1			
			M	L	C
Q.1	**a.**	Determine the complement of the following function: (i) $F = xy' + x'y$ (ii) $F = x'yz' + x'y'z$	06	L3	CO1
	b.	Describe map method for three variables.	04	L2	CO1
	c.	Apply K map technique to simplify the following function: (i) $F(x, y, z) = \Sigma(0, 2, 4, 5, 6)$ (ii) $F(x, y, z) = x'y + yz' + y'z'$	10	L3	CO1
		OR			
Q.2	**a.**	Apply K map technique to simplify the function : $F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$ and $d(w, x, y, z) = \Sigma(0, 2, 5)$	06	L3	CO1
	b.	Determine all the prime implicants for the Boolean function F and also determine which are essential $F(w, x, y, z) = \Sigma(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$	10	L3	CO1
	c.	Develop a verilog gate-level description of the circuit shown in Fig.Q2(c).  Fig.Q2(c)	04	L3	CO1
		Module – 2			
Q.3	**a.**	Explain the combinational circuit design procedure with code conversion example.	10	L2	CO2
	b.	Design a full adder circuit. Also develop data flow verilog model for full adder.	10	L3	CO2
		OR			
Q.4	**a.**	Describe 4 × 1 MUX with block diagram and truth table. Also develop a behavioral model verilog code for 4 × 1 MUX.	10	L2	CO2
	b.	What are storage elements? Explain the working of SR and D latch along with logic diagram and function table.	10	L2	CO2
		Module – 3			
Q.5	**a.**	Explain the basic operational concepts between the processor and memory.	10	L2	CO3
	b.	Describe the following: (i) Processor clock (ii) Basic performance equation (iii) Clock rate (iv) SPEC rating	10	L2	CO3
		OR			
Q.6	**a.**	Define addressing mode. Explain any four types of addressing mode with example.	10	L2	CO3

	b.	Mention four types of operations to be performed by instructions in a computer. Explain the basic types of instruction formats to carry out. $C \leftarrow [A] + [B]$	10	L2	CO3
		Module – 4			
Q.7	a.	With a neat diagram, explain the concept of accessing I/O devices.	10	L2	CO4
	b.	What is bus arbitration? Explain centralized and distributed arbitration method with a neat diagram.	10	L2	CO4
		OR			
Q.8	a.	With neat sketches, explain various methods for handling multiple interrupts requests raised by multiple devices.	10	L2	CO4
	b.	What is cache memory? Explain any two mapping function of cache memory.	10	L2	CO4
		Module – 5			
Q.9	a.	Draw the single bus architecture and write the control sequence for execution of instruction ADD (R_3), R_1.	10	L3	CO5
	b.	With suitable diagram, explain the concept of register transfer and fetching of word from memory.	10	L2	CO5
		OR			
Q.10	a.	With a neat diagram, explain the flow of 4-stage pipeline operation.	10	L2	CO5
	b.	Explain the role of cache memory and pipeline performance.	10	L2	CO5

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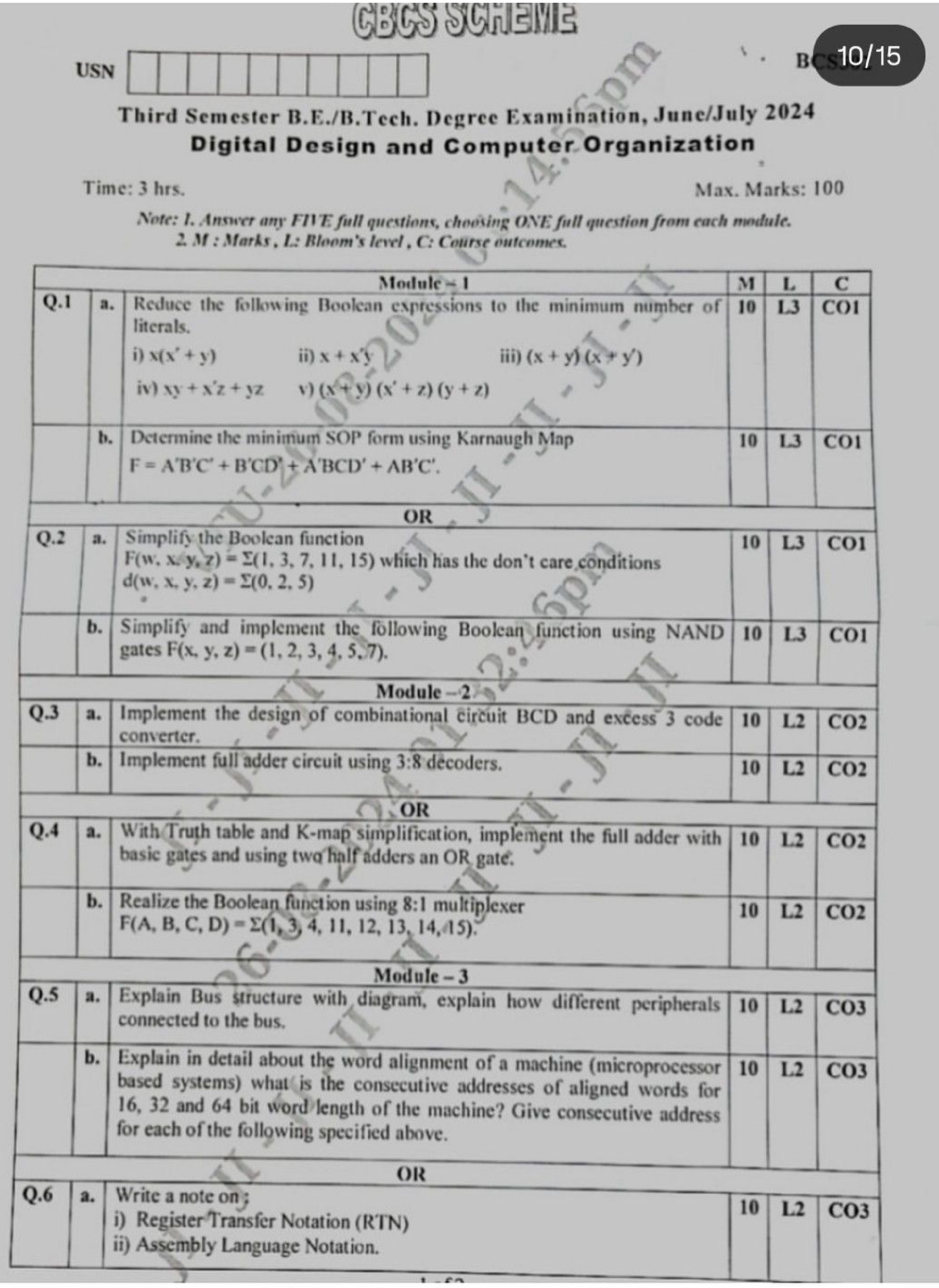


CBCS SCHEME

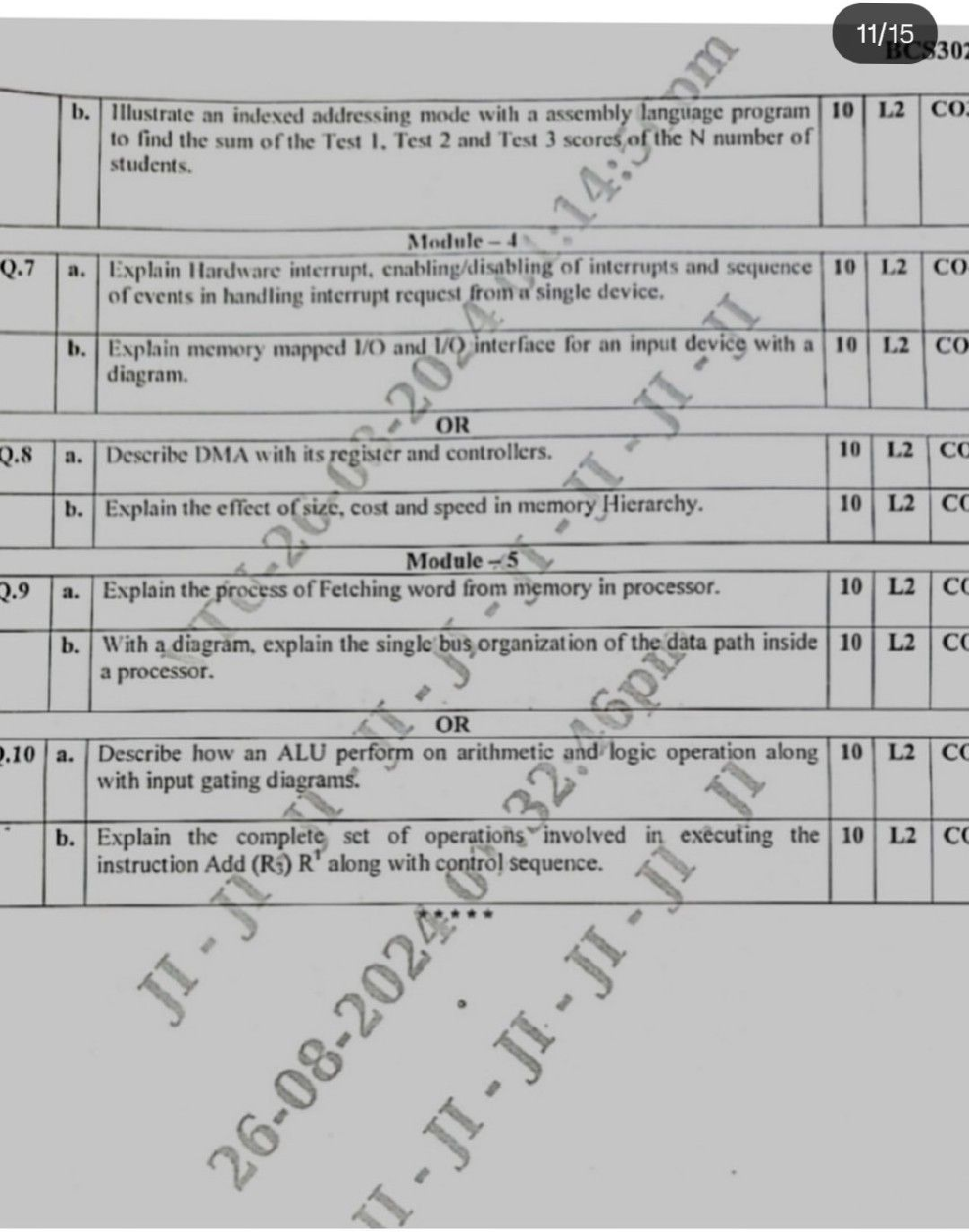
USN ☐☐☐☐☐☐☐☐☐☐ B

Third Semester B.E./B.Tech. Degree Examination, June/July 2024
Digital Design and Computer Organization

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Reduce the following Boolean expressions to the minimum number of literals. i) $x(x' + y)$ ii) $x + x'y$ iii) $(x + y)(x + y')$ iv) $xy + x'z + yz$ v) $(x + y)(x' + z)(y + z)$	10	L3	CO1
	b.	Determine the minimum SOP form using Karnaugh Map $F = A'B'C' + B'CD' + A'BCD' + AB'C'$.	10	L3	CO1
		OR			
Q.2	a.	Simplify the Boolean function $F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$ which has the don't care conditions $d(w, x, y, z) = \Sigma(0, 2, 5)$	10	L3	CO1
	b.	Simplify and implement the following Boolean function using NAND gates $F(x, y, z) = (1, 2, 3, 4, 5, 7)$.	10	L3	CO1
		Module – 2			
Q.3	a.	Implement the design of combinational circuit BCD and excess 3 code converter.	10	L2	CO2
	b.	Implement full adder circuit using 3:8 decoders.	10	L2	CO2
		OR			
Q.4	a.	With Truth table and K-map simplification, implement the full adder with basic gates and using two half adders an OR gate.	10	L2	CO2
	b.	Realize the Boolean function using 8:1 multiplexer $F(A, B, C, D) = \Sigma(1, 3, 4, 11, 12, 13, 14, 15)$.	10	L2	CO2
		Module – 3			
Q.5	a.	Explain Bus structure with diagram, explain how different peripherals connected to the bus.	10	L2	CO3
	b.	Explain in detail about the word alignment of a machine (microprocessor based systems) what is the consecutive addresses of aligned words for 16, 32 and 64 bit word length of the machine? Give consecutive address for each of the following specified above.	10	L2	CO3
		OR			
Q.6	a.	Write a note on ; i) Register Transfer Notation (RTN) ii) Assembly Language Notation.	10	L2	CO3



	b.	Illustrate an indexed addressing mode with a assembly language program to find the sum of the Test 1, Test 2 and Test 3 scores of the N number of students.	10	L2	CO

Module – 4

Q.7	a.	Explain Hardware interrupt, enabling/disabling of interrupts and sequence of events in handling interrupt request from a single device.	10	L2	CO
	b.	Explain memory mapped I/O and I/O interface for an input device with a diagram.	10	L2	CO

OR

Q.8	a.	Describe DMA with its register and controllers.	10	L2	CO
	b.	Explain the effect of size, cost and speed in memory Hierarchy.	10	L2	CO

Module – 5

Q.9	a.	Explain the process of Fetching word from memory in processor.	10	L2	CO
	b.	With a diagram, explain the single bus organization of the data path inside a processor.	10	L2	CO

OR

.10	a.	Describe how an ALU perform on arithmetic and logic operation along with input gating diagrams.	10	L2	CO
	b.	Explain the complete set of operations involved in executing the instruction Add (R_3) R^1 along with control sequence.	10	L2	CO

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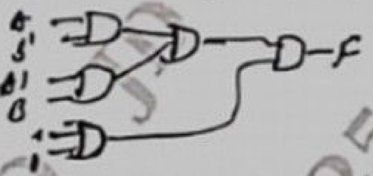
26-08-2024

Third Semester B.E./B.Tech. Degree Supplementary Examination
June/July 2024
Digital Design and Computer Organization

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Simplify the Boolean function i) $F(x, y, z) = \Sigma(2, 3, 4, 5)$ ii) $F(x, y, z) = \Sigma(3, 4, 6, 7)$	10	L3	CO1
	b.	Obtain a minimum product of sum with a Karnaugh Map $F(w, x, y, z) = x'z' + wyz + w'y'z' + x'y$	10	L3	CO1
		OR			
Q.2	a.	Define multiplexer. Explain 2 to 1 line multiplexer.	10	L2	CO1
	b.	Write the verilog code and time diagram for the given circuit with propagation delay where the AND, OR gate has a delay of 30ns and 10ns.  Fig Q2(b)	5	L2	CO1
	c.	Explain implementation of full adder with logic diagram.	5	L3	CO1
		Module – 2			
Q.3	a.	Explain with neat diagram and 4 input priority encodes.	10	L2	CO2
	b.	Explain 2 : 4 time decoder with help of logic diagram and truth table.	10	L2	CO2
		OR			
Q.4	a.	Define Latch. Explain S-R flip flop based on NOR Gate with neat diagram.	10	L2	CO2
	b.	Explain clocked D flip flop with neat diagram.	10	L2	CO2
		Module – 3			
Q.5	a.	With neat diagram, explain the basic operational concepts of computers.	10	L2	CO3
	b.	Write a program to evaluate arithmetic statement $Y = (A + B) \cdot (C + D)$ using 3 address, 2 address, one address and zero address instruction.	10	L3	CO3
		OR			
Q.6	a.	Describe the concept of Blanch instruction with example.	10	L2	CO3
	b.	Explain 5 addressing modes with example.	10	L2	CO3

Module – 4

Q.7	a.	Explain the I/O interfacing and I/O device with computers.
	b.	What is Bus Arbitration? Explain types of bus arbitration.

OR

Q.8	a.	What is cache memory? Explain the different type of cache mappi function.
	b.	Explain basic concepts involved for memory structures of computers.

Module – 5

Q.9	a.	Explain with neat diagram of single bus organization.
	b.	Explain complete execution steps for instruction ADD (R3), R1.

OR

Q.10	a.	Explain execution of complete instruction carry out.
	b.	What is pipeline? Explain with example of pipeline performance.

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Note: 01. Answer any FIVE full questions, choosing at least ONE question from each

		Module -1	*Bl Tax L
Q.01	a	Demonstrate the nonassociativity of the NOR gate	
	b	Design a car safety alarm circuit diagram. The system considers four inputs: door (D), key (K), seat pressure (P) and seat belt (B). The input is considered HIGH (1) if the door is closed, the key is in, the driver is on the seat, or the seat belt is fastened. The alarm (A) should sound with two conditions as stated below: The door is not closed, and the key is in. The door is closed, the key is in the driver's seat, and the seat belt is not closed. (a) Construct a truth table for the system based on input arrangement D, K, P, B with A as an output (b)Design a Karnaugh map to verify the simplified expression (c) Draw the simplified circuit using NOR gates only	
	c	With an example explain the working of Test Bench in Verilog.	
		OR	
Q.02	a	Demonstrate the positive and negative logic signal	
	b	A digital system is to be designed in which the month of the year is given as I/P in four-bit form. The month of January is represented as '0000', February as "0001" and so on. The output of the system should correspond to the input of the month containing 31 days, or otherwise, it is '0'. Consider the excess number in the I/P beyond 1011' as don't care condition: (i) Write truth table, SOP Em and POSIIM form (ii) Simplify for SOP using K-map (iii) Realize using basic gates	
	c	What is User-Defined Primitives in Verilog? What are the general rules for UDP? Explain with an example HDL for user defined primitive. Draw the Schematic for the Circuit with UDP_02467	
		Module-2	
Q. 03	a	Differentiate Latches and Flip-Flops.	
	b	Explain the working of Four-bit adders using 4-full Adders.	
	c	Implement Y (A, B, C, D) = $\sum$m (0, 1, 6, 7, 8, 9, 10, 11, 12, 14) using 16-to-1 multiplexer and 8-to-1 multiplexer.	
		OR	
Q.04	a	Explain different modeling styles used to write the code in VERILOG with an example.	
	b	Design a BCD-to-excess-3 code converter.	

	c	Define decoder. Describe the working principle of a 3:8 decoder. Draw the logic diagram of the 3:8 decoder with enabled input. Realize the following Boolean expressions using a 3:8 decoder and multi-input OR gates: F1(A, B, C) = ∑ m(1, 3, 7) F2(A, B, C) = ∑m(2, 3, 5).
		Module-3
Q. 05	a	Write one address, two address, and three address instructions to carry out C← [A] + [B].
	b	Explain the basic operation concepts of the computer with a neat diagram.
	c	Explain a) Processor Clock, b) Basic performance equation, c) Clock rate d) Performance measurement
		OR
Q. 06	a	Write ALP of adding a list of n numbers using indirect addressing mode
	b	What is an addressing mode? Explain the different addressing modes. With an example for each.
	c	Explain the following: (i) Byte addressability (ii) Big-endian assignment (i) Little-endian assignment.
		Module-4
Q. 07	a	Draw a neat block diagram of memory hierarchy in a computer system. Discuss the variation of size, speed and cost per bit in the hierarchy.
	b	What is DMA Bus arbitration? Explain different bus arbitration techniques.
		OR
Q. 08	a	With neat sketches, explain various methods for handling multiple interrupt requests raised by multiple devices.
	b	What is cache memory? Explain the different mapping functions used in cache memory.
		Module-5
Q. 09	a	Explain the single-bus organization of computers and fundamental concepts with a neat diagram.
	b	Write and explain the control sequence for execution of the instruction Add(R3), R1
		OR
Q. 10	a	Explain with an example the different types of hazards that occur during pipelining.
	b	Write and explain the control sequence for the execution of an unconditional branch instruction.

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs attained by every bit of question.